Varendra University

Dept. of Computer Science & Engineering (CSE)

Multiplication and Division Instructions (Chapter-9)

Course Title: **Microprocessor and Assembly Language**
Course No: **CSE 3205**

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Outline

1. MUL and IMUL.
2. Simple Applications of MUL and IMUL.
3. DIV and IDIV.
4. Sign Extension of the Dividend.

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MUL and IMUL

MUL (multiply) for unsigned multiplication.
IMUL (integer multiply) for signed multiplication.

The syntax is,

```
MUL    source
and
IMUL   source
```

MUL and IMUL

Byte Form:

For byte multiplication, one number is contained in the source and the other is assumed to be in AL. The 16-bit product will be in AX.

Word Form:

For word multiplication, one number is contained in the source and the other is assumed to be in AX.

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Simple Applications of MUL and IMUL

Example 9.6: Translate the high-level language assignment statement A = 5 × A - 12 × B into assembly code. Let A and B be word variables, and suppose there is no overflow. Use IMUL for multiplication.

Solution:

```
MOV   AX,5          ;AX = 5
IMUL  A             ;AX = 5 x A
MOV   A,AX          ;A = 5 x A
MOV   AX,12         ;AX = 12
IMUL  B             ;AX = 12 x B
SUB   A,AX          ;A = 5 x A - 12 x B
```

Simple Applications of MUL and IMUL

Example 9.7: Write a procedure FACTORIAL that will compute N! for a positive integer N. The procedure should receive N in CX and return N! in AX. Suppose that overflow does not occur.

Solution:

```
FACTORIAL       PROC
;computes N!
;input: CX = N
;output: AX = N!
                MOV   AX,1          ;AX holds product
TOP:
                MUL   CX            ;product = product x term
                LOOP  TOP
                RET
FACTORIAL       ENDP
```

CX is bot

DIV and IDIV

When division is performed, we obtain two results,

1) Quotient.
2) Remainder.

- DIV (divide) is used for unsigned division.
- IDIV (integer divide) for signed division.

The syntax is,

```
DIV divisor
and
IDIV divisor
```

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DIV and IDIV

Byte Form:

- In this form, the divisor is an 8-bit register or memory byte.
- The 16-bit dividend is assumed to be in AX.
- After division, the 8-bit quotient is in AL and the 8-bit remainder Is in AH.

Word Form:

- Here the divisor is a 16-bit register or memory word.
- The 32-bit dividend is assumed to be in DX:AX.
- After division, the 16-bit quotient is in AX and the 16-bit remainder is in DX.

Sign Extension of the Dividend

Example 9.12: Divide -1250 by 7.

Solution:

```
MOV   AX,-1250          ;AX gets dividend
CWD                     ;Extend sign to DX
MOV   BX,7              ;BX has divisor
IDIV BX                 ;AX gets quotient, DX has remainder
```

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Sign Extension of the Dividend

Example 9.13: Divide the signed value of the byte variable XBYTE by – 7.

Solution:

```
MOV   AL,XBYTE          ;AL has dividend
CBW                     ;Extend sign to AH
MOV   BL,-7             ;BL has divisor
IDIV  BL                ;AL has quotient, AH has remainder
```

There is no effect of CBW and CWD on the flag.

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